

AKILAN MANIKANDAN

Backend & Automation Engineer | Python | FastAPI | RPA | Applied AI Systems

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SUMMARY

Backend and automation engineer with 0-1 year of internship experience building production-grade API services, workflow automation pipelines, and data ingestion systems. Proficient in Python, Java, TypeScript, and SQL, with hands-on delivery across FastAPI, Spring Boot, Django REST Framework, NestJS, PostgreSQL, and MongoDB. Demonstrated ability to own features end-to-end — from schema design and API implementation to deployment and monitoring. Adept at converting repetitive operational processes into reliable, configuration-driven systems with full audit trails.

TECHNICAL SKILLS

Languages	Python, Java, TypeScript, JavaScript, SQL, C++
Backend	FastAPI, Spring Boot, Flask, Django REST Framework, NestJS, Next.js, React
Automation	Playwright, PyAutoGUI, Selenium, TagUI, Scrapy, n8n
Databases	PostgreSQL, MongoDB, Mongoose, SQLAlchemy, psycopg2
Data / ETL	Pandas, NumPy, Power BI REST API
Cloud / Storage	AWS, boto3
Auth / Security	JWT, Passport, bcrypt, AES, RSA, ABAC, OTP, MFA, TOTP
ML / AI	TensorFlow, Keras, Librosa, scikit-learn, LangChain, OpenAI API, Gemini API
Tools	Git, Maven, MS Office

INTERSHIPS

Software Engineer | *Clarivium Technologies*

Aug 2025 – Mar 2026

- Engineered **50+** Python automation scripts converting recurring report-generation and file-handling processes into reusable, **configuration-driven** workflows — eliminating an estimated 8+ hours of daily manual effort across operations teams.
- Built end-to-end RPA pipelines using **Playwright** and **PyAutoGUI** covering login, OTP/TOTP prompts, date filtering, report export, retry logic, and cleanup — automating **12 recurring workflows** with zero manual intervention on success paths.
- Designed a **PostgreSQL**-backed execution orchestration layer with idempotent state tracking, date-range derivation, standardized file naming, and audit logging — reducing duplicate pipeline runs by **~90%** and enabling full replayability of failed stages.
- Developed **pandas**- and **SQLAlchemy**-based ingestion handlers for **5 dataset types** (sales, purchase, inventory, service, enquiry) with CSV sanitization, branch-specific transformations, and delete-and-reload semantics — maintaining data consistency across **20+ daily ingestion cycles**.
- Automated output delivery by uploading renamed report files to **AWS S3** using **boto3**, deriving file names from PostgreSQL metadata — removing manual file organization and ensuring consistent downstream availability.
- Built and maintained **48** documented **n8n** workflows (21 branchwise KPI refresh + 25 dashboard freshness monitoring) integrating **Power BI** executeQueries API, **PostgreSQL** upserts, and notification steps — replacing manual analytics validation for a 10-branch automotive client.
- Built **15+ Mautic-n8n automation workflows** on **AWS EC2** for **Raya MVP**, a dealership campaign product, automating **WhatsApp** and **email outreach** via **AiSensy BSP APIs** across use cases including **insurance renewals**, **service reminders**, and **test-drive follow-ups** — reducing campaign setup time **from hours to minutes**.
- Designed and implemented **branch-level and client-specific contact segmentation pipelines** serving **500+ dealership contacts** across multiple **tenant configurations**, enabling **personalized campaign delivery** at scale without manual list management.

- Engineered the migration from manual campaign operations to a **Mautic-n8n automation backend**, cutting recurring campaign execution effort significantly and enabling the team to manage outreach across **3–5 dealership branches** from a **single orchestrated workflow layer**.

Backend Developer | [CleandeskAI](#)

Apr 2025 – Jul 2025

- Designed and shipped 5 **Django REST Framework** API endpoints supporting document classification and workspace management features in a production PostgreSQL-backed environment.
- Diagnosed and resolved **10+** backend defects across API response handling and data model inconsistencies, improving endpoint reliability and reducing client-reported errors.
- Implemented backend data model improvements and query fixes in **PostgreSQL**, improving data consistency and application stability for active users.

Backend Engineer | [Simbiotik Technologies](#)

May 2024 – Jun 2024

- Built backend modules for an HRMS platform using **NestJS**, **MongoDB**, and **Mongoose** — implementing CRUD APIs for users, employees, tasks, and policy management with role-guarded endpoints.
- Implemented **JWT**- and **Passport**-based authentication with **bcrypt** password hashing, OTP-based MFA, and **ABAC** role guards — securing all API routes against unauthorized access.
- Added document upload handling with **randomized filenames**, **MIME-type validation**, and size limits, supporting document-driven employee onboarding workflows.

AI/ML Engineer | [Zidio Development](#)

Jul 2024 – Aug 2024

- Built a speech emotion recognition pipeline on the TESS dataset using **TensorFlow/Keras** — implementing **MFCC** feature extraction with **Librosa** and an **LSTM** sequence model for 8-class emotion classification.
- Achieved classification accuracy improvement through iterative hyperparameter tuning and sequence padding strategies across training and validation splits.

PROJECTS

Secure Data Storage & Controlled File Sharing System | [Github Repo](#)

Tech Stack: Java, Spring Boot, Flask, AES-256, RSA, ABAC, scikit-learn (Isolation Forest)

- Architected a multi-role secure file-sharing system in **Spring Boot** (25+ Java source files) supporting upload, access-request, approval, deletion, and download workflows with full audit trail across **5 distinct user roles**.
- Implemented **AES-256** file encryption, **RSA** key exchange for secure key distribution, and policy-based **ABAC** access control — enforcing least-privilege access throughout the full file lifecycle.
- Built a **Flask** anomaly-detection microservice using an **Isolation Forest** model and rule-based scoring to flag suspicious access behavior and trigger real-time administrator email alerts.
- Added **OTP-based authentication**, policy persistence, and access-request tracking — improving security enforcement auditability and operational traceability.

CityPulse — Event Recommendation Chatbot | [Github Repo](#)

Tech Stack: FastAPI, Next.js, React, MongoDB, LangChain, OpenAI GPT, Scrapy, Selenium

- Built a **FastAPI** backend and **Next.js** frontend for an event recommendation chatbot with Google OAuth sign-in, MongoDB session persistence, and **LangChain**-powered conversational flows delivering recommendations in **under 2 seconds** average response time.
- Engineered a **LangChain RAG** pipeline with conversation memory, keyword and timeframe entity extraction, and preference tracking over a **1,200+ event** MongoDB corpus — generating structured recommendations with contextual relevance.
- Developed a **Scrapy** + **Selenium** scraping pipeline covering **17 Chennai event categories** — normalizing date/time fields and ingesting structured records into MongoDB for real-time chatbot retrieval.

CERTIFICATIONS

Claude Code in Action	Anthropic	Mar 2026
AI Fluency for Students	Anthropic	Mar 2026
NLP for ML with Python: Advanced NLP Using spaCy & Scikit-learn	Skillsoft	Apr 2025
Python for Data Science	NPTEL	Apr 2025
DBMS Completion	Udemy	Apr 2024

EDUCATION

B.Tech — Computer Science and Engineering	2022 – 2026
SRM Institute of Science and Technology	
Lalaji Memorial Omega International School	2020 – 2022
CLASS XII (CBSE)	
Hindu Senior Secondary School	2019 – 2020
CLASS X (CBSE)	